ARM64 HPC Research Topics

Graduate Thesis Topics in HPC and Data Analytics

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May 17, 2026

1. Energy Efficiency of ARM64 for Scientific HPC Workloads

Research Question

Under which workload characteristics does ARM64 deliver better performance-per-watt than x86-64 for scientific HPC applications?

Abstract

This research evaluates whether modern ARM64 processors provide better energy efficiency than x86-64 systems for scientific HPC workloads. The study benchmarks numerical and parallel applications to analyze runtime, power consumption, thermal behavior, and scalability.

Tooling

LINPACK, HPCG, Linux perf, ARM Streamline, OpenMP

References

- F. Mantovani et al. "Performance and Energy Consumption of HPC Workloads on a Cluster Based on ARM ThunderX2 CPUs." *FGCS*, 2020.
- N. Simakov et al. "Are We Ready for Broader Adoption of ARM in the HPC Community?" *HPCAsia Workshops*, 2023.

2. Memory-Centric Profiling on ARM HPC Systems

Research Question
Can ARM SPE-based memory profiling identify optimization opportunities that traditional CPU profiling misses?

Abstract
This thesis investigates whether memory-centric profiling techniques can identify bottlenecks and improve optimization strategies for ARM-based HPC systems. The research focuses on cache utilization, memory locality, bandwidth saturation, and profiling-guided optimization for scientific applications.

Tooling
perf, ARM SPE, PAPI, STREAM benchmark

References

- Samuel Miksits, Johannes Weidendorfer, and Dieter Kranzlmuller. "Multi-level Memory-Centric Profiling on ARM Processors with ARM Statistical Profiling Extension (SPE)." *arXiv preprint arXiv:2410.01514*, 2024.
- Arm Ltd. "Statistical Profiling Extension." Arm Developer Documentation and Architecture Blog Series, 2023–2024.

3. Transformer Inference on ARM Many-Core CPUs

Research Question

What software and hardware factors determine transformer inference throughput and latency on ARM many-core CPUs?

Abstract

This research investigates transformer inference performance on ARM many-core CPUs. The study analyzes thread scheduling, memory usage, vectorization efficiency, quantization, and runtime optimization techniques for modern AI workloads.

Tooling

PyTorch, ONNX Runtime, Hugging Face Transformers, perf

References

- Canqun Jiang, Zheng Wang, Xiaoyang Wang, and David Kaeli. "Full-Stack Optimizing Transformer Inference on ARM Many-Core CPUs." *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 9, pp. 2478–2491, 2023.
- Canqun Jiang, Zheng Wang, and David Kaeli. "Characterizing and Optimizing Transformer Inference on ARM Many-Core Processors." *Proceedings of the International Workshop on Performance, Portability and Productivity in HPC*, 2022.

4. AI-Assisted Autotuning for ARM HPC Workloads

Research Question

Can machine-learning-based autotuning improve performance and energy efficiency on ARM HPC systems?

Abstract

This thesis explores the use of machine learning and Bayesian optimization techniques for automatic tuning of HPC workloads on ARM architectures. The research investigates parameter optimization, compiler tuning, scheduling decisions, and performance prediction for scientific applications.

Tooling

OpenTuner, GPTune, Python, LLVM/Clang, perf

References

- Yurii Oleynik, Jack Deslippe, and Samuel Williams. “GPTune: Multitask Learning for Autotuning HPC Applications.” *IEEE Transactions on Parallel and Distributed Systems*, vol. 32, no. 9, pp. 2229–2245, 2021.
- Nikolay Simakov, Hartwig Anzt, and Thomas Papatheodore. “Are We Ready for Broader Adoption of ARM in the HPC Community?” *Proceedings of the ACM International Conference on High Performance Computing in Asia-Pacific Workshops*, 2023.

5. Memory-Bound Parallel Workloads on ARM64 and x86-64

Research Question

How do ARM64 and x86-64 architectures differ in cache efficiency, memory bandwidth saturation, and OpenMP scaling for memory-bound scientific workloads?

Abstract

This thesis investigates memory-bound scientific applications on ARM64 and x86-64 systems. The study evaluates cache efficiency, memory bandwidth utilization, NUMA behavior, and OpenMP scalability to identify architectural differences that influence performance and energy efficiency.

Tooling

STREAM, OpenMP, Linux perf, PAPI, LIKWID

References

- Filippo Mantovani, Edoardo Calore, Alberto Schenck, and Thomas Fahringer. "Performance and Energy Consumption of HPC Workloads on a Cluster Based on ARM ThunderX2 CPUs." *Future Generation Computer Systems*, vol. 103, pp. 126–138, 2020.
- Zhen Shi, Hartwig Anzt, and Jack Dongarra. "ARM SVE Unleashed: Performance and Insights Across HPC Applications on NVIDIA Grace." *arXiv preprint arXiv:2505.09462*, 2025.

Conclusion

Key Research Themes

- Energy-efficient high performance computing
- Memory-centric optimization and profiling
- AI inference on ARM many-core processors
- Machine-learning-assisted performance tuning
- Memory-bound scientific application analysis

Why ARM64?

- Rapid growth in HPC and cloud infrastructure
- Improved performance-per-watt characteristics
- Increasing adoption in AI and scientific computing
- Strong opportunities for systems analytics research